

Mitochondrial Dysfunction and Chronic Liver Disease.

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Mitochondria are generally considered the powerhouse of the cell, a small subcellular organelle that produces most of the cellular energy in the form of adenosine triphosphate (ATP). In addition, mitochondria are involved in various biological functions, such as biosynthesis, lipid metabolism, oxidative phosphorylation, cell signal transduction, and apoptosis. Mitochondrial dysfunction is manifested in different aspects, like increased mitochondrial reactive oxygen species (ROS), mitochondrial DNA (mtDNA) damage, adenosine triphosphate (ATP) synthesis disorder, abnormal mitophagy, as well as changes in mitochondrial morphology and structure. Mitochondrial dysfunction is related to the occurrence and development of various chronic liver diseases, including hepatocellular carcinoma (HCC), viral hepatitis, drug-induced liver injury (DILI), alcoholic fatty liver (AFL), and non-alcoholic fatty liver (NAFL). In this review, we summarize and discuss the role and mechanisms of mitochondrial dysfunction in chronic liver disease, focusing on and discussing some of the latest studies on mitochondria and chronic liver disease.

Shifting patterns of cellular energy production (adenosine triphosphate) over the day and key timings for the effect of optical manipulation.

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Mitochondria are optically responsive organelles producing energy for cell function via adenosine triphosphate (ATP). But ATP production appears to vary over the day. Here we use *Drosophila melanogaster* to reveal daily shifts in whole animal ATP production in a tight 24 hours' time series. We show a marked production peak in the morning that declines around midday and remains low through afternoon and night. ATP production can be improved with long wavelengths (>660 nm), but apparently not at all times. Hence, we treated flies with 670 nm light to reveal optimum times. Exposures at 670 nm resulted in a significant ATP increases and a shift in the ATP/adenosine diphosphate (ADP) ratio at 8.00 and 11.00, whilst application at other time points had no effect. Hence, light-induced ATP increases appear limited to periods when natural production is high. In summary, long wavelength influences on mitochondria are conserved across species from fly to human. Determining times for their administration to improve function in ageing and disease are of key importance. This study progresses this problem.

Targeted Treatment Strategies for Mitochondria Dysfunction: Correlation with Neurological Disorders.

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Mitochondria are an essential intracellular organelle for medication targeting and delivery since they seem to create energy and conduct many other cellular tasks, and mitochondrial dysfunctions and malfunctions lead to many illnesses. Many initiatives have been taken to detect, diagnose, and image mitochondrial abnormalities, and to transport and accumulate medicines precisely to mitochondria, all because of special mitochondrial aspects of the pathophysiology of cancer. In addition to the negative membrane potential and paradoxical mitochondrial dynamics, they include high temperatures, high levels of reactive oxygen species, high levels of glutathione, and high temperatures. Neurodegenerative

diseases represent a broad spectrum of debilitating illnesses. They are linked to the loss of certain groups of neurons based on an individual's physiology or anatomy. The mitochondria in a cell are generally accepted as the authority with respect to ATP production. Disruption of this system is linked to several cellular physiological issues. The development of neurodegenerative disorders has been linked to mitochondrial malfunction, according to pathophysiological studies. There seems to be substantial evidence connecting mitochondrial dysfunction and oxidative stress to the development of neurodegenerative disorders. It has been extensively observed that mitochondrial malfunction triggers autophagy, which plays a role in neurodegenerative disorders. In addition, excitotoxicity and mitochondrial dysfunction have been linked to the development of neurodegenerative disorders. The pathophysiology of neurodegenerative illnesses has been linked to increased apoptosis and necrosis, as well as mitochondrial malfunction. A variety of synthetic and natural treatments have shown efficacy in treating neurodegenerative illnesses caused by mitochondrial failure. Neurodegenerative illnesses can be effectively treated with existing drugs that target mitochondria, although their precise formulations are poorly understood. Therefore, there is an immediate need to focus on creating drug delivery methods specifically targeted at mitochondria in the treatment and diagnosis of neurodegenerative disorders.

Tying trafficking to fusion and fission at the mighty mitochondria.

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The mitochondrion is a unique organelle that serves as the main site of ATP generation needed for energy in the cell. However, mitochondria also play essential roles in cell death through apoptosis and necrosis, as well as a variety of crucial functions related to stress regulation, autophagy, lipid synthesis and calcium storage. There is a growing appreciation that mitochondrial function is regulated by the dynamics of its membrane fusion and fission; longer, fused mitochondria are optimal for ATP generation, whereas fission of mitochondria facilitates mitophagy and cell division. Despite the significance of mitochondrial homeostasis for such crucial cellular events, the intricate regulation of mitochondrial fusion and fission is only partially understood. Until very recently, only a single mitochondrial fission protein had been identified. Moreover, only now have researchers turned to address the upstream machinery that regulates mitochondrial fusion and fission proteins. Herein, we review the known GTPases involved in mitochondrial fusion and fission, but also highlight recent studies that address the mechanisms by which these GTPases are regulated. In particular, we draw attention to a substantial new body of literature linking endocytic regulatory proteins, such as the retromer VPS35 cargo selection complex subunit, to mitochondrial homeostasis. These recent studies suggest that relationships and cross-regulation between endocytic and mitochondrial pathways may be more widespread than previously assumed.

Mitochondrial transcription and translation: overview.

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Mitochondria are the major source of ATP in the cell. Five multi-subunit complexes in the inner membrane of the organelle are involved in the oxidative phosphorylation required for ATP production. Thirteen subunits of these complexes are encoded by the mitochondrial genome often referred to as mtDNA. For this reason, the expression of mtDNA is vital for the assembly and functioning of the oxidative phosphorylation complexes. Defects of the mechanisms regulating mtDNA gene expression have been associated with deficiencies in assembly of these complexes, resulting in mitochondrial diseases.

Recently, numerous factors involved in these processes have been identified and characterized leading to a deeper understanding of the mechanisms that underlie mitochondrial diseases.

Mitochondrial Damage due to Hypoxia and Its Forensic Significance.

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Mitochondria are the special organelle in eukaryotic cells. Their main functions are to synthesize energy required for cell activity by oxidative phosphorylation. Most of the oxygen absorbed by the body is consumed in the mitochondria. The precise diagnosis of mechanical asphyxia is one of the difficulties in forensic pathology practice. Forensic pathologists have been trying to find a reliable and sensitive marker for the diagnosis of mechanical asphyxia. Mitochondria are very sensitive to hypoxic environments, and the markers of mitochondrion damage can be used as a basis for the diagnosis of mechanical asphyxia. The purpose of this paper is to review the research progress on mitochondrial damage in hypoxic environments and to explore the possibility of using markers of mitochondrion damage in forensic pathological practice.

Mitochondrial function in development and disease.

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Mitochondria are organelles with vital functions in almost all eukaryotic cells. Often described as the cellular 'powerhouses' due to their essential role in aerobic oxidative phosphorylation, mitochondria perform many other essential functions beyond energy production. As signaling organelles, mitochondria communicate with the nucleus and other organelles to help maintain cellular homeostasis, allow cellular adaptation to diverse stresses, and help steer cell fate decisions during development. Mitochondria have taken center stage in the research of normal and pathological processes, including normal tissue homeostasis and metabolism, neurodegeneration, immunity and infectious diseases. The central role that mitochondria assume within cells is evidenced by the broad impact of mitochondrial diseases, caused by defects in either mitochondrial or nuclear genes encoding for mitochondrial proteins, on different organ systems. In this Review, we will provide the reader with a foundation of the mitochondrial 'hardware', the mitochondrion itself, with its specific dynamics, quality control mechanisms and cross-organelle communication, including its roles as a driver of an innate immune response, all with a focus on development, disease and aging. We will further discuss how mitochondrial DNA is inherited, how its mutation affects cell and organismal fitness, and current therapeutic approaches for mitochondrial diseases in both model organisms and humans.

Quantitative proteomic comparison of rat mitochondria from muscle, heart, and liver.

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Mitochondria, through oxidative phosphorylation, are the primary source of energy production in all tissues under aerobic conditions. Although critical to life, energy production is not the only function of mitochondria, and the composition of this organelle is tailored to meet the specific needs of each cell

type. As an organelle, the mitochondrion has been a popular subject for proteomic analysis, but quantitative proteomic methods have yet to be applied to tease apart subtle differences among mitochondria from different tissues or muscle types. Here we used mass spectrometry-based proteomics to analyze mitochondrial proteins extracted from rat skeletal muscle, heart, and liver tissues. Based on 689 proteins identified with high confidence, mitochondria from the different tissues are qualitatively quite similar. However, striking differences emerged from the quantitative comparison of protein abundance between the tissues. Furthermore we applied similar methods to analyze mitochondrial matrix and intermembrane space proteins extracted from the same mitochondrial source, providing evidence for the submitochondrial localization of a number of proteins in skeletal muscle and liver. Several proteins not previously thought to reside in mitochondria were identified, and their presence in this organelle was confirmed by protein correlation profiling. Hierarchical clustering of microarray expression data provided further evidence that some of the novel mitochondrial candidates identified in the proteomic survey might be associated with mitochondria. These data reveal several important distinctions between mitochondrial and submitochondrial proteomes from skeletal muscle, heart, and liver tissue sources. Indeed approximately one-third of the proteins identified in the soluble fractions are associated predominantly to one of the three tissues, indicating a tissue-dependent regulation of mitochondrial proteins. Furthermore a small percentage of the mitochondrial proteome is unique to each tissue.

Mitochondrial dynamics in cancer and neurodegenerative and neuroinflammatory diseases.

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Mitochondria are key organelles in the cell, hosting essential functions, from biosynthetic and metabolic pathways, to oxidative phosphorylation and ATP production, from calcium buffering to red-ox homeostasis and apoptotic signalling pathways. Mitochondria are also dynamic organelles, continuously fusing and dividing, and their localization, size and trafficking are finely regulated. Moreover, in recent decades, alterations in mitochondrial function and dynamics have been implicated in an increasing number of diseases. In this review, we focus on the relationship clarified hitherto between mitochondrial dynamics and cancer, neurodegenerative and neuroinflammatory diseases.

How to get rid of mitochondria: crosstalk and regulation of multiple mitophagy pathways.

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Mitochondria are indispensable cellular organelles providing ATP and numerous other essential metabolites to ensure cell survival. Reactive oxygen species (ROS), which are formed as side reactions during oxidative phosphorylation or by external agents, induce molecular damage in mitochondrial proteins, lipids/membranes and DNA. To cope with this and other sorts of organellar stress, a multi-level quality control system exists to maintain cellular homeostasis. One critical level of mitochondrial quality control is the removal of damaged mitochondria by mitophagy. This process utilizes parts of the general autophagy machinery, e.g. for the formation of autophagosomes but also employs mitophagy-specific factors. Depending on the proteins utilized mitophagy is divided into receptor-mediated and

ubiquitin-mediated mitophagy. So far, at least seven receptor proteins are known to be required for mitophagy under different experimental conditions. In contrast to receptor-mediated pathways, the Pink-Parkin-dependent pathway is currently the best characterized ubiquitin-mediated pathway. Recently two additional ubiquitin-mediated pathways with distinctive similarities and differences were unraveled. We will summarize the current state of knowledge about these multiple pathways, explain their mechanism, and describe the regulation and crosstalk between these pathways. Finally, we will review recent evidence for the evolutionary conservation of ubiquitin-mediated mitophagy pathways.
